2

INVERTER ACTIVITY

Build an Inverter

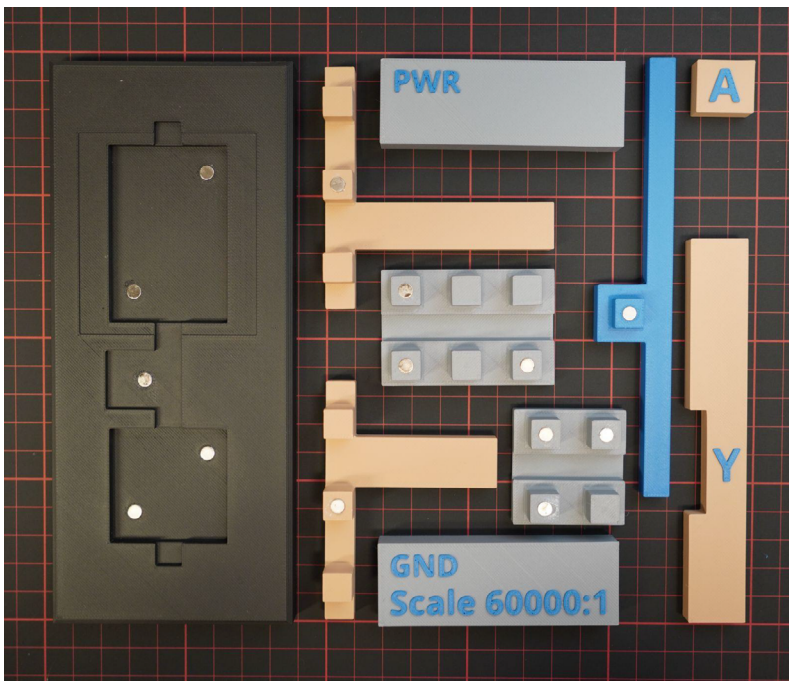
Computer chips work by chaining together billions of simple logic gates. The simplest logic gate is an inverter - it converts a 0 to a 1 and vice versa.

A transistor is an electronic switch with no moving parts. The first inverters used a single transistor and a resistor.

CMOS inverters were invented in the 1960s as a way to save energy. They use two transistors, one to pull the output low, and the other to pull the output high.

The MOS in CMOS stands for **MOSFET**, which is a type of transistor that can be easily made with **photolithography.** The C stands for complementary, because the N and P type transistors work together.

In this exercise you'll build a scale model of a CMOS inverter.

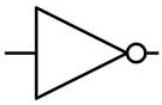


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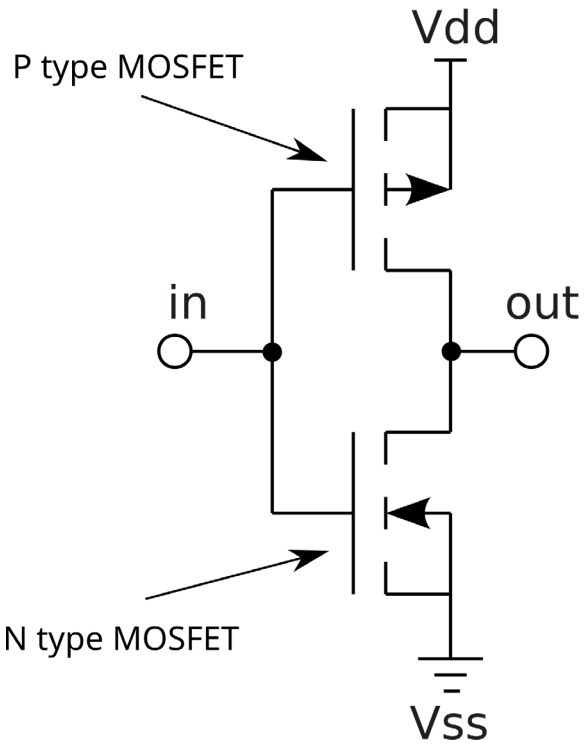
2

INVERTER ACTIVITY

Inverter



Input	Output
0	1
1	0



- Can you fit all the pieces together?

  The model uses A for in and Y for out as that's traditional and easier to fit on the model!
- Can you see how the schematic on the left matches the physical model?
- This is a scale model, the real version that's inside one of our chips is 60,000 times smaller!
- Why do you think the P MOSFET is larger than the N MOSFET?

Scan the code for

- More information about the way the inverter works.
- Files for printing your own model or a link for buying a kit.

tinytapeout.com/inverter-model

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